

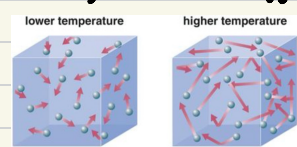
# LEC11 - Energy & Conservation

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## Learning Goals:

- Define momentum and angular momentum and discuss situations in which energy is exchanged and conserved
- Define several types of energy (kinetic, potential, thermal) and describe situations in which energy is exchanged and conserved
- Relate conservation of energy and momentum to Newton's laws, which we discussed last time

## \* Example of Thermal Energy Density & Kinetic Energy



- \* Many forms of energy such as thermal energy do not have formulas to calculate them

- \* The formula for gravitational potential energy is  $GPE = \text{mass} \times \text{gravity} \times \text{height}$ , but it **DOES NOT** hold for objects in orbit

## Conservation Laws:

### Momentum

- **Momentum = Mass  $\times$  Velocity**
- A net force changes momentum, usually by causing an acceleration (change in velocity)
- Conservation of momentum is another way of phrasing Newton's first and second laws
- When a planet orbits a star, the planet's momentum changes (velocity direction changes), however the momentum of the star also changes, so momentum is conserved
- Because momentum is conserved, both objects in a system actually orbit around their center of mass
- Each object traces its own (elliptical) orbit around the center of mass
- We use the "wobble" of stars to find exoplanets

### Angular Momentum

- **Angular momentum = radius  $\times$  mass  $\times$  velocity**
- The angular momentum of an object cannot change unless an external twisting force (torque) is acting on it
- In orbits, angular momentum is conserved
- Angular momentum is why objects rotate faster when they shrink in radius
- Torque is a force applied at some radius perpendicular to the rotational axis
- Torque is applied at some radius perpendicular to the rotational axis
- Ex. to make a door rotate, you must apply a torque at the far end of the hinge (bigger radius)
- Kepler's second law is a consequence of conservation of angular momentum
- **Angular momentum =  $r \times mv$** 
  - $\rightarrow r \times mv$  stays the same for the whole orbit
  - $\rightarrow \times$  means cross product, but you don't need to know what that is

### Energy

- 1. Kinetic Energy (based on motion)**
  - Thermal energy is a type of kinetic energy
  - Depends on speed (magnitude of velocity)
  - **$K = \frac{1}{2}mv^2$** 
    - Kinetic Energy (J)
    - Mass (kg)
    - Speed (m/s)
  - Speed makes biggest difference
- 2. Potential Energy (Stored energy)**
  - In space, an object or gas cloud has more gravitational energy when it is spread out than when it contracts
  - A contracting cloud converts gravitational potential energy to thermal energy
  - Energy can be converted between potential & kinetic energy (falling objects)
  - The total sum of kinetic gravitational potential energy is called the Total Mechanical energy
  - Total mechanical energy is conserved for bound planetary orbits
- 3. Thermal Energy (Kinetic Energy of all particles)**
  - Temperature measures the average kinetic energy of many particles of a substance
  - Temperature depends on kinetic energy **AND** density
  - Mechanical Energy can be converted to thermal energy
    - $\rightarrow$  Ex. Satellite heating up when entering atmosphere due to drag
- 4. Radiative Energy (Light)**
  - Will be covered in another lecture

- \* Many different things are conserved in bound planetary orbits, including momentum, angular momentum, and energy
- \* Momentum is the inertia of an object calculated by  $P = m \times v$ . In an orbit, the momentum of a planet and its star change individually, but as a system their sum is constant
- \* Angular momentum is the angular inertia of an object calculated by  $P_A = r \times mv$ . Angular momentum is caused by torque
- \* Energy has many forms, such as kinetic, potential, mechanical, and thermal. Energy can freely transition between these forms under specific circumstances